EFFECTIVE HERMITIAN–YANG–MILLS FLOW ON NODAL DEGENERATIONS OF CURVES

BERND JOHANNES WUEBBEN

Abstract. We prove a finite-dimensional limit for Hermitian–Yang–Mills metric flow on the trivial bundle over a compact curve approaching a reduced nodal fibre. The total space is smooth and Kähler, the central components are smooth, and each node joins distinct components. On the time scale $\log(1/|t|)$, where t is the degeneration parameter, the limiting component metrics follow the gradient flow of the sum of their squared symmetric-space distances, with component areas as masses and one coupling for each node. For uniformly bounded determinant-one initial metrics with prescribed local L^2 limits away from the nodes, we prove uniform global L^2 tracking on bounded rescaled time intervals and pointwise tracking away from the nodes after a fixed physical-time transient. The construction combines component Green functions, exact stationary neck metrics and the Donaldson trace inequality. At the constant flat metric, we also identify the rescaled linearized heat evolution on the small spectral subspace and prove a uniform gap on its orthogonal complement. A rank-two example on a smoothing of a triangle of projective lines has component metrics that admit no simultaneous diagonalization by a fixed holomorphic frame.

1. Introduction

When a compact curve separates into several components, the scalar Laplacian acquires small eigenvalues associated with functions that are nearly constant on each component. For a bundle with parallel endomorphisms, the same phenomenon occurs in the linearization of Hermitian–Yang–Mills metric flow. We ask whether the evolving component metrics admit a finite-dimensional description with an error tending to zero as the curve degenerates.

We prove such a description for a trivial holomorphic bundle. Its finite variables are positive Hermitian matrices, coupled across each node by their Riemannian logarithm. Component areas supply the masses. The equation is nonlinear even for the trivial bundle: metrics on three components need not admit a simultaneous diagonalization. The comparison holds for actual HYM solutions on intervals whose physical length grows like the logarithm of the inverse smoothing parameter.

1.1. **The family and finite equation.** Let $\pi : \mathcal{X} \to \Delta$ be a proper holomorphic map from a smooth complex surface to a disc, with connected fibres. Assume that $\mathcal{X}$ carries a smooth Kähler form ω, that $X_t = \pi^{-1}(t)$ is smooth for $t \neq 0$, and that $X_0 = C_1 \cup \cdots \cup C_N$ is reduced, with smooth irreducible components and only ordinary nodes. Each node joins two distinct components. Let $\mathcal{E}$ be the set of nodes, regarded as edges of the component graph, retaining

Date: November 10, 2026
This version: November 10, 2026.
2020 *Mathematics Subject Classification.* 53C07 (primary); 58J35, 14D06, 58J50 (secondary).
Key words and phrases. Hermitian–Yang–Mills flow, nodal curve, degeneration, matrix metric, small eigenvalue, finite-dimensional approximation.

repeated edges. Put

$$A_i = \int_{C_i} \omega > 0, \qquad L = L(t) = \log(1/|t|).$$

All statements are uniform in the argument of sufficiently small t. The induced metric on X_t stays fixed during each flow.

Fix the trivial holomorphic rank-r bundle, $r \geq 2$, with its global frame and determinant metric one. Write

$$\mathcal{P}_r = \{H = H^* > 0 : \det H = 1\}.$$

Its symmetric-space metric and distance are

$$g_H(U,V) = \operatorname{tr}(H^{-1}UH^{-1}V), \qquad d(H,K)^2 = \operatorname{tr}\big(\log(H^{-1/2}KH^{-1/2})^2\big). \tag{1}$$

The principal matrix logarithm $\log(H^{-1}K)$ is self-adjoint for H; it need not be Hermitian in the fixed frame.

Given $\mathbf{H}^0 \in \mathcal{P}_r^N$, define $\mathbf{H}(\tau)$ by

$$A_i H_i^{-1} \frac{dH_i}{d\tau} = 2\pi \sum_{p:\, i \leftrightarrow j} \log(H_i^{-1}H_j), \qquad H_i(0) = H_i^0. \tag{2}$$

There is one summand for each incident node. This is the negative gradient flow of

$$\mathscr{E}(\mathbf{H}) = \pi \sum_{p:\, i \leftrightarrow j} d(H_i, H_j)^2 \tag{3}$$

for the product metric with weights A_i, each edge occurring once in the energy. It has a global solution in a compact subset of $\mathcal{P}_r^N$ (Lemma 2.1). For later comparison put

$$\sigma(H,K) = \operatorname{tr}(H^{-1}K) + \operatorname{tr}(K^{-1}H) - 2r \geq d(H,K)^2. \tag{4}$$

1.2. **Actual-flow tracking.** Our HYM convention is

$$H^{-1}\partial_s H = -2i\Lambda_{\omega_t} F_H, \qquad F_H = \bar\partial(H^{-1}\partial H). \tag{5}$$

The degree and central curvature constant are zero. The variable s is physical time and $\tau = s/L$ is slow time. For compact $K_i \Subset C_i \setminus \operatorname{Sing} X_0$, let $K_i(t) \subset X_t$ be its image under the identifications of Lemma 3.1. All global integrals use $dA_t = \omega|_{X_t}$.

Theorem 1.1. *For every initial tuple $\mathbf{H}^0$ there are smooth determinant-one metrics $\widehat{H}_t(s)$ on the trivial bundle with the following properties. For every fixed $T < \infty$, they and their inverses are uniformly bounded for $0 \leq s \leq TL$, and*

$$\sup_{0 \leq s \leq TL} \sup_{K_i(t)} d(\widehat{H}_t(s), H_i(s/L)) \leq C_{K_i,T}/L. \tag{6}$$

Let H_t^{in} be smooth determinant-one metrics satisfying $cI \leq H_t^{\mathrm{in}} \leq CI$ with constants independent of t, and assume that

$$\eta_t = \int_{X_t} \sigma(H_t^{\mathrm{in}}, \widehat{H}_t(0))\, dA_t \longrightarrow 0.$$

The actual HYM solution $H_t(s)$ exists smoothly for all $s \geq 0$, and

$$\sup_{0 \leq s \leq TL} \int_{X_t} d(H_t(s), \widehat{H}_t(s))^2\, dA_t \leq \eta_t + C_T \frac{(1+\log L)^2}{L}. \tag{7}$$

For each fixed compact K_i one also has

$$\sup_{1\le s\le TL}\sup_{K_i(t)} d(H_t(s),H_i(s/L))^2 \le C_{K_i,T}\left(\eta_t+\frac{(1+\log L)^2}{L}\right). \tag{8}$$

The constants may depend on the fixed family, rank, initial tuple, c, C and T, but not on t or the spatial derivatives of the initial metrics. The interval in (8) is nonempty for small t.

The comparison metrics are constructed from component Green functions, matrix exponentials and geodesics. In particular the initial class is nonempty: take $H_t^{\mathrm{in}}=\widehat{H}_t(0)$. Uniform bounds and L^2 convergence of the initial metrics to H_i^0 on every fixed compact component subset also imply $\eta_t\to 0$, by the node-area bound. Initial derivatives and bounded initial neck values are unrestricted.

The global estimate has no waiting time. The pointwise estimate has a fixed physical-time transient, equal to $1/L$ in slow time. It concerns fixed compact subsets away from the nodes and makes no assertion of uniform pointwise tracking through the shrinking neck. The theorem also does not assert tracking uniform as $T\to\infty$.

1.3. **Linearized evolution and a compact example.** At the constant identity metric, (5) linearizes to the scalar heat equation on trace-free Hermitian matrix entries. Write Q for the component graph Laplacian, $A=\mathrm{diag}(A_i)$ and $B=2\pi A^{-1}Q$. The scalar theorem of [Wue26d] gives $N-1$ small positive eigenvalues and a complementary gap. Theorem 6.1 identifies the low space by component averages and proves convergence of the rescaled heat evolution to $e^{-\tau B}$ in operator norm. Tensoring gives $(N-1)(r^2-1)$ small positive modes and r^2-1 exact parallel modes. The finite matrix flow has the same linearization at a constant tuple.

The cubic family $xyz=t(x^3+y^3+z^3)$ has a central triangle with component areas one in the normalized Fubini–Study metric. The tuple

$$I,\qquad \begin{pmatrix}2&0\\0&1/2\end{pmatrix},\qquad \begin{pmatrix}5/3&4/3\\4/3&5/3\end{pmatrix} \tag{9}$$

cannot be simultaneously diagonalized by a fixed complex congruence. Section 7 verifies the geometry and the genuinely noncommuting initial data. Its rank-two positive linearized cluster has six modes, each asymptotic to $6\pi/L$.

1.4. **Predecessors and proof.** Donaldson's heat flow and metric comparison provide the analytic framework [Don85, HKKP18]. Haiden–Katzarkov–Kontsevich–Pandit describe algebraically determined asymptotics on a fixed curve, and Lam treats broader fixed compact Kähler backgrounds under his stated hypotheses [Lam26]. The local spectral and phase comparisons in [Wue26a, Wue26b] also concern fixed smooth backgrounds; their regularity constants cannot be assumed uniform under a nodal degeneration.

Daskalopoulos–Wentworth already study actual Yang–Mills flow near the boundary of Teichmüller space [DW00]. Their uniform long-time argument uses a positive adjoint spectral gap when the limiting representations on the normalization components are irreducible; their spectral analysis also detects small adjoint eigenvalues. The componentwise parallel endomorphisms of the trivial bundle lie outside that gap regime. The present theorem computes their finite nonlinear evolution on logarithmic time and reconstructs actual flows.

The intrinsic squared-distance graph energy and its logarithmic gradient are established constructions [TAV13]. The assertion here is their occurrence with these geometric masses and capacities in an HYM degeneration. The HYM equation contains a commutator term

absent from ordinary symmetric-space harmonic-map heat flow (12). We retain that term and use exact stationary one-variable profiles only on the necks.

Harmonic-map degeneration has a separate compactness theory. Zhu [Zhu10] proves energy identities and compactness criteria for bounded-energy sequences of harmonic maps from degenerating hyperbolic surfaces to a compact target. Rupflin–Topping–Zhu [RTZ13] study Teichmüller harmonic map flow, in which both the map and its hyperbolic domain metric evolve; for global solutions degenerating only at infinite time, they obtain subsequential limits given by branched minimal immersions or constant maps on compactified components. Here each induced fibre metric remains fixed during its HYM flow, and the comparison is uniform on bounded logarithmic time intervals. The commutator in (12) also distinguishes the two evolution equations.

The proof adds Green-function corrections to the component metrics and matches them to geodesic neck profiles. The resulting curvature residual has space-time L^1 norm tending to zero on $0 \le s \le TL$. A forced Donaldson trace inequality proves global tracking, and local scalar parabolic estimates give pointwise convergence away from the nodes. The argument does not require a global unweighted Schauder estimate uniform under degeneration. The separate spectral comparison provides uniform Hilbert-space inverse and heat bounds on the complement.

2. Matrix metrics and flow comparison

2.1. **The finite flow.** The space $\mathcal{P}_r$ is a complete simply connected manifold of nonpositive sectional curvature. The geodesic from P to Q is

$$\mathrm{Geo}(P,Q;\theta) = P^{1/2}\exp\bigl(\theta\log(P^{-1/2}QP^{-1/2})\bigr)P^{1/2}. \tag{10}$$

Its initial velocity is $\mathrm{Log}_P Q = P\log(P^{-1}Q)$. These formulas and the distance (1) are standard [Bha13, Sections 2.3–2.6]. The determinant-one space is totally geodesic: the logarithm of the determinant along (10) is the affine interpolation of the endpoint logarithmic determinants. Squared distance is jointly convex along pairs of geodesics, and every two points have a unique geodesic. All maps in (10) are smooth, including at repeated eigenvalues.

Lemma 2.1. *Equation (2) is the negative gradient flow of (3) for the product metric with weights A_i. It preserves determinant one and exists for all $\tau \ge 0$ in a compact subset depending only on its initial tuple and the areas. All its derivatives of each fixed order are bounded on bounded time intervals.*

Proof. The gradient of $\frac{1}{2}d(P,Q)^2$ in the first variable is $-\mathrm{Log}_P Q$; applying this identity at each vertex gives (2), including the factor 2π and the weight A_i^{-1}. The same intrinsic graph-gradient formula appears in [TAV13, Section III]. Since $\operatorname{tr}\log(H_i^{-1}H_j) = 0$, determinant one is preserved.

Let $o = (I,\dots,I)$ in the product, and write d_A for its weighted distance. Geodesic convexity and $\mathscr{E}(o) = 0$ give

$$\langle \nabla\mathscr{E}(\mathbf{H}), \mathrm{Log}_{\mathbf{H}}\, o\rangle_A \le -\mathscr{E}(\mathbf{H}).$$

Along the negative gradient flow this is

$$\frac{d}{d\tau}\frac{1}{2}d_A(\mathbf{H},o)^2 \le -\mathscr{E}(\mathbf{H}) \le 0.$$

Closed bounded sets in this finite-dimensional complete manifold are compact. The smooth vector field therefore continues for all time, and differentiation of the ordinary differential equation gives the stated derivative bounds. □

2.2. **The exact HYM equation.** In a holomorphic coordinate $z = x + iy$, write $\omega_t = (i/2)a(z)\,dz \wedge d\bar{z}$. The nonnegative real scalar Laplacian is $\Delta_t = -4a^{-1}\partial_z\partial_{\bar{z}}$. Equation (5) reads

$$\partial_s H = 4a^{-1}(H_{z\bar{z}} - H_{\bar{z}}H^{-1}H_z). \tag{11}$$

Indeed $F_H = -\partial_{\bar{z}}(H^{-1}H_z)\,dz \wedge d\bar{z}$. Expanding the complex derivatives gives

$$\begin{aligned} a\partial_s H = {} & H_{xx} + H_{yy} - H_x H^{-1}H_x - H_y H^{-1}H_y \\ & + i(H_x H^{-1}H_y - H_y H^{-1}H_x). \end{aligned} \tag{12}$$

The final term is Hermitian. It is generally nonzero and is absent from ordinary symmetric-space harmonic-map heat flow. For example, take $H(x, y) = \exp(xU + yV)$ with noncommuting trace-free Hermitian matrices U, V. At the origin the first four terms cancel, whereas the last is $i[U, V]$. The proof below makes no assumption that this term vanishes for an actual solution.

In rank one $H = e^u$ gives $\partial_s u = -\Delta_t u$. Taking the trace of the logarithmic velocity in general gives the same heat equation for $\log\det H$, so the determinant normalization persists. Linearization at a constant identity matrix is consequently $\partial_s u = -\Delta_t u$ on trace-free Hermitian entries.

2.3. **A comparison estimate with a residual.** For a smooth comparison family $K(s)$ define

$$R(K) = K^{-1}\partial_s K + 2i\Lambda_{\omega_t}F_K. \tag{13}$$

It is self-adjoint with respect to K. Its pointwise norm below is the operator norm in that metric.

Lemma 2.2. *Let $H(s)$ be an exact HYM solution on a fixed compact fibre, and $K(s)$ any smooth positive comparison family. With $h = H^{-1}K$,*

$$(\partial_s + \Delta_t)\sigma(H, K) \le \operatorname{tr}((h - h^{-1})R(K)). \tag{14}$$

If $mI \le H, K \le MI$ throughout the interval, then

$$\int_{X_t} \sigma(H(s), K(s))\,dA_t \le \int_{X_t} \sigma(H(0), K(0))\,dA_t + \frac{2rM}{m}\int_0^s \int_{X_t} |R(K)|\,dA_t\,du. \tag{15}$$

The constants do not use a spectral gap or a Sobolev constant of X_t.

Proof. For two exact flows, the Donaldson trace calculation gives $(\partial_s + \Delta_t)\sigma \le 0$; see also [Wan08, Definition 2.3 and Proposition 2.4], with the bundle morphism zero. We recall the pointwise calculation to fix conventions. Choose a time-independent holomorphic frame normal for H at the point, with $H = I$ and $\partial H = 0$. The curvature terms cancel between the time derivative and the scalar Laplacian of $\operatorname{tr}(H^{-1}K)$, leaving

$$-4a^{-1}\operatorname{tr}(K_{\bar{z}}K^{-1}K_z) \le 0.$$

This is nonpositive because $K_{\bar{z}} = K_z^*$. Interchanging H, K proves the same assertion for the inverse trace.

Relative to the exact comparison velocity, the actual velocity of K has the additional term $KR(K)$. Its contribution to the two trace derivatives is respectively $\operatorname{tr}(hR)$ and $-\operatorname{tr}(Rh^{-1})$. This proves (14). Both h and R are K-self-adjoint, since $Kh = KH^{-1}K$ is Hermitian. Therefore

$$|\operatorname{tr}((h - h^{-1})R)| \le (\operatorname{tr} h + \operatorname{tr} h^{-1})|R| \le 2r(M/m)|R|.$$

Integration over the whole compact fibre removes the Laplacian; the metric and area form are fixed during the flow. This gives (15). □

If e^{b_j} are the eigenvalues of $H^{-1}K$, then

$$d(H,K)^2 = \sum_j b_j^2 \leq \sum_j (e^{b_j} + e^{-b_j} - 2) = \sigma(H,K),$$

which proves the inequality in (4).

Lemma 2.3. *For every fixed smooth compact fibre and smooth positive initial metric, the HYM metric flow exists smoothly for all $s \geq 0$. For the trivial holomorphic bundle, initial bounds $cI \leq H(0) \leq CI$ imply the same bounds at every later time. These comparison constants are independent of the degeneration parameter.*

Proof. The first assertion is the classical compact-base existence theorem for Donaldson's heat flow [Don85]; the convention (5) and the all-time statement are also given in [HKKP18, Section 4.1, equation (4.8) and following paragraph]. The order-preservation principle [HKKP18, Proposition 4.3] compares the solution with the two stationary constant matrix metrics cI, CI. Neither comparison changes as the base parameter varies. Determinant one follows from the scalar heat equation for $\log\det H$. □

3. Plumbing geometry and component potentials

The smooth total-space and nodal-fibre assumptions give holomorphic coordinates u, v at each node in which $\pi = uv$. Choose disjoint such bidiscs and a fixed small outer radius $r_* > 0$. On the two central branches, u and v are puncture coordinates. We use the same coordinates to normalize the logarithmic parts of Green functions.

Lemma 3.1. *There are identifications of the thick part of X_t with the components punctured at radius ρ, for $|t| \ll \rho^2$ and $0 < \rho \ll r_*$, having the following properties.*

1. *In the shrinking branch collars they are exactly $u \mapsto (u, t/u)$ or $v \mapsto (t/v, v)$. On a fixed exterior they depend smoothly on the real and imaginary parts of t and equal the identity at $t = 0$.*
2. *In each fixed derivative order k, pulled-back metric and complex-structure coefficients differ from the central ones by at most $C_k|t|\rho^{-n_k}$ for some fixed integer n_k.*
3. *The middle part of each neck, where $|u|, |v| \leq 2\rho$, has area at most $C\rho^2$. Each transition annulus $|u| \asymp \rho$ has area $O(\rho^2)$ and metric uniformly comparable to that of the central branch.*

All constants are uniform in the argument of t.

Proof. Remove fixed small node discs from the central components. The map π is a submersion on a compact neighbourhood of the resulting exterior, including its boundary. Choose smooth horizontal lifts of the two real parameter directions. On a collar of each boundary, choose the lifts that keep u, or respectively v, fixed. A partition of unity interpolates these lifts only on fixed annuli, where all coefficients are smooth and bounded. Flow along the straight segment from zero to t. This gives a smooth trivialization of the fixed exterior which agrees with the coordinate identification on smaller boundary collars. Extend it inward by $u \mapsto (u, t/u)$ on each branch. The two branch images are disjoint when $|t| \ll \rho^2$.

On the fixed exterior all coefficient differences are $O_k(|t|)$. On the shrinking collar,

$$|D^k(t/u)| \leq C_k|t|\rho^{-k-1}.$$

The induced metric is the restriction of a smooth ambient tensor along this map. Differentiating it gives a finite power of ρ^{-1} in every fixed order. The complex coordinate on the collar is exactly u, so its complex structure needs no correction there. The smooth branch metric is nondegenerate through the puncture, and the same estimates hold for inverse coefficients.

The ambient metric is bounded above by a fixed multiple of the flat metric in the bidisc. In the parameter u, its fibre area is bounded by

$$C \int_{|t|/(2\rho)\le|u|\le 2\rho} (1 + |t|^2/|u|^4)\, dx\, dy \le C'\rho^2.$$

On $|u| \asymp \rho$ the second term is uniformly small, and the induced metric is comparable to the branch metric. The estimates involve only $|t|$. □

We shall take

$$\rho = L^{-1}. \tag{16}$$

Every geometric coefficient error required below is then bounded by $Ce^{-L}L^n$, and is smaller than any negative power of L. Fixed compact subsets are transported by the exterior identifications from Lemma 3.1.

Lemma 3.2. *For each node puncture $p \in C_i$, there is a real mean-zero function $g_{i,p}$, smooth off p, such that*

$$\Delta_i g_{i,p} = -2\pi\delta_p + \frac{2\pi}{A_i}, \qquad \int_{C_i} g_{i,p}\, dA_i = 0. \tag{17}$$

In the assigned puncture coordinate u, it has the form $g_{i,p} = \log|u| + \gamma_{i,p}(u)$, with $\gamma_{i,p}$ smooth.

Proof. For the nonnegative Laplacian, $\Delta \log|u| = -2\pi\delta_0$. Multiply the logarithm by a cutoff equal to one near the puncture. Its Laplacian is the stated point mass plus a smooth function. The right side of (17) has integral zero, so the difference is a smooth mean-zero function. Invert the scalar Laplacian on the mean-zero functions of the fixed compact smooth component and adjust the additive constant. Elliptic regularity makes the correction smooth. This also proves uniqueness. □

Along the finite trajectory let $B_{ij}(\tau) = \log(H_i(\tau)^{-1}H_j(\tau))$, and define

$$\xi_i(\tau, z) = -\sum_{p:\, i\leftrightarrow j} B_{ij}(\tau) g_{i,p}(z), \qquad V_i = H_i^{-1}\partial_\tau H_i. \tag{18}$$

These matrices are H_i-self-adjoint and trace free. The essential compatibility condition is

$$\Delta_i \xi_i = -V_i \qquad \text{away from the punctures.} \tag{19}$$

Indeed (17) gives the left side as $-(2\pi/A_i)\sum B_{ij}$, exactly the negative of (2).

For $\tau \in [0, T]$, the finite trajectory and its derivatives are bounded. Consequently, away from discs of radius ρ,

$$\begin{gathered} |\xi_i| \le C_T(1 + |\log\rho|), \qquad |\nabla\xi_i| \le C_T(1 + r^{-1}), \\ |\nabla^2\xi_i| \le C_T(1 + r^{-2}), \qquad \int |\xi_i|\, dA_i \le C_T, \qquad \int |\nabla\xi_i|^2\, dA_i \le C_T(1 + |\log\rho|), \end{gathered} \tag{20}$$

where r is distance in a puncture chart when near a puncture. The same bounds hold after any fixed number of slow-time derivatives. They follow by integrating $|\log r|$ and r^{-2} against the smooth component area. All norms of matrices in this proof are equivalent on the compact range of the component metrics.

4. Construction of an approximate trajectory

Throughout this section $L = \log(1/|t|)$, $\rho = L^{-1}$ and $\tau = s/L$. We construct a smooth global metric by matching corrected component metrics to exact stationary neck profiles. The endpoint matrices move in slow time; the resulting neck profile therefore has a small time residual, although its curvature vanishes exactly.

4.1. **Component metrics.** On a component punctured at radius $\rho/2$, put

$$K_i(\tau, z) = H_i(\tau) \exp(\xi_i(\tau, z)/L). \tag{21}$$

It is positive Hermitian with determinant one. By (20), $|\xi_i|/L = O_T(\log L/L)$ there. Transfer its matrix entries to the corresponding part of X_t using Lemma 3.1.

Lemma 4.1. *On the transferred component regions,*

$$\int |R(K_i)|\, dA_t \le C_T \frac{1 + \log L}{L^2} + C_T e^{-L} L^n. \tag{22}$$

On every fixed compact subset away from the punctures, the residual and its fixed finite spatial derivatives are $O_T(L^{-2})$.

Proof. First calculate on the central component. Write $h = \xi_i/L$ and suppress the component index. For bounded small h,

$$e^{-h}\partial e^h = \int_0^1 e^{-ah}(\partial h)e^{ah}\, da = \Phi(\mathrm{ad}_h)\partial h,$$

where Φ is smooth, $\Phi(0) = I$, and its derivatives are bounded. After applying $\bar\partial$ and contracting, the difference from the linear term is bounded by

$$|2i\Lambda F_{He^h} - \Delta h| \le C_T(|h||\Delta h| + |\nabla h|^2).$$

All commutators remain in this expression. The reference matrix H is spatially constant. For the time derivative,

$$(He^h)^{-1}\partial_s(He^h) = L^{-1}e^{-h}Ve^h + e^{-h}\partial_s e^h,$$

whose difference from V/L is bounded by $C_T L^{-2}(|\xi||V| + |\partial_\tau \xi|)$. Thus

$$R(K_i) = L^{-1}(V_i + \Delta_i \xi_i) + O_T\big(L^{-2}\big(|\xi_i||V_i| + |\partial_\tau \xi_i| + |\nabla \xi_i|^2 + |\xi_i||\Delta_i \xi_i|\big)\big)\,. \tag{23}$$

The first term vanishes by (19). Integration of the other terms uses (20) and proves the first term of (22). The coordinate and coefficient errors in Lemma 3.1, multiplied by the polynomial derivative bounds of the potentials, contribute $O_T(e^{-L}L^n)$. On a fixed compact subset every potential derivative is bounded, which proves the local assertion. □

4.2. **Geodesic necks.** At a node p joining C_i, C_j, write

$$\xi_i = -B_{ij}\log|u| + c_{i,p} + q_{i,p}(u), \qquad q_{i,p}(0) = 0, \tag{24}$$

where the regular part is smooth, with bounded slow-time derivatives. Use the analogous expansion on the v branch. Set

$$P_{i,p} = H_i \exp(c_{i,p}/L), \qquad P_{j,p} = H_j \exp(c_{j,p}/L),$$

and on $uv = t$ define

$$N_p(\tau, u) = \mathrm{Geo}(P_{i,p}, P_{j,p}; \theta), \qquad \theta = -\frac{\log|u|}{L}. \tag{25}$$

Since $1 - \theta = -\log|v|/L$, the same expression from the opposite end is obtained by reversing the geodesic. In the neck region used below, $0 \le \theta \le 1$.

Lemma 4.2. *For every fixed τ, the metric N_p has zero Chern curvature. On $|u|, |v| \le 2\rho$ its time residual satisfies*

$$|R(N_p)| \le C_T/L, \qquad \int |R(N_p)|\, dA_t \le C_T \rho^2/L. \tag{26}$$

Proof. Put $S = \log(P_{i,p}^{-1/2} P_{j,p} P_{i,p}^{-1/2})$. The spatial connection is exactly

$$N_p^{-1} \partial N_p = P_{i,p}^{-1/2} S P_{i,p}^{1/2}\, \partial\theta.$$

The coefficient matrix is spatially constant and θ is harmonic, so its $\bar\partial$ derivative is zero. The matrices $P_{i,p}$ and their slow-time derivatives stay in bounded sets, and geodesic interpolation is smooth on their compact range. Thus $|N_p^{-1}\partial_s N_p| \le C_T/L$. The area estimate is Lemma 3.1(3). □

4.3. **Differentiated matching.** Near the i end use the logarithmic matrix coordinate $\ell_i(K) = \log(H_i^{-1} K)$. It takes values in the space of H_i-self-adjoint trace-free endomorphisms. All its derivatives needed here are bounded on the compact matrix range.

Lemma 4.3. *On the transition annulus $\rho \le |u| \le 2\rho$, for $k = 0, 1, 2$,*

$$|D^k(\ell_i(K_i) - \ell_i(N_p))| \le C_T \left(\frac{\rho^{1-k}}{L} + \frac{1 + |\log \rho|}{L^2 \rho^k} \right). \tag{27}$$

The same estimate holds after any fixed number of slow-time derivatives. The corresponding assertion holds at the other end.

Proof. Write $\varepsilon = L^{-1}$ and, at a fixed slow time, consider

$$F(\varepsilon, \theta) = \ell_i(\mathrm{Geo}(H_i e^{\varepsilon c_{i,p}}, H_j e^{\varepsilon c_{j,p}}; \theta)).$$

Its derivatives are uniformly bounded for $\theta \in [0,1]$ and small ε. Exactly, $F(0,\theta) = \theta B_{ij}$ and $F(\varepsilon, 0) = \varepsilon c_{i,p}$. The difference from their sum vanishes on both coordinate axes. The double integral of its mixed derivative therefore gives

$$F(\varepsilon, \theta) = \theta B_{ij} + \varepsilon c_{i,p} + \varepsilon\theta G(\varepsilon, \theta, \tau),$$

with G and its required derivatives uniformly bounded. On the annulus, $\theta = -\varepsilon \log|u|$ and $D^k\theta = O(\varepsilon \rho^{-k})$ for $k = 1, 2$. The component logarithm is

$$\ell_i(K_i) = \theta B_{ij} + \varepsilon c_{i,p} + \varepsilon q_{i,p}(u).$$

Since $q_{i,p}$ is smooth and zero at the puncture, $|D^k q_{i,p}| \le C_T \rho^{1-k}$ for $k \le 2$, using a harmless weaker bound when $k = 2$. Subtraction and the chain rule give (27). The same calculation after slow-time differentiation uses the bounded derivatives of the finite flow. No pair of the matrices has been assumed to commute. □

Choose a cutoff $\chi(|u|/\rho)$ equal to zero near the inner end and one near the outer end of the transition annulus. Set there

$$\widehat{H}_t = H_i \exp(\ell_i(N_p) + \chi(\ell_i(K_i) - \ell_i(N_p))). \tag{28}$$

Use K_i on the thick pieces and N_p on the middle necks. The formula is positive Hermitian with determinant one and agrees with the adjacent metrics on neighbourhoods of the boundaries. It thus defines a smooth global metric. Positivity and uniform two-sided bounds follow from compactness of the component tuple, smallness of ξ_i/L on the thick pieces, and compactness of the neck geodesics.

Proposition 4.4. *For every fixed $T < \infty$, the constructed metrics satisfy*

$$\sup_{0\le s\le TL}\int_{X_t}|R(\widehat{H}_t(s))|\,dA_t \le C_T\frac{(1+\log L)^2}{L^2}, \tag{29}$$

and consequently

$$\int_0^{TL}\int_{X_t}|R(\widehat{H}_t(s))|\,dA_t\,ds \le C_T\frac{(1+\log L)^2}{L}. \tag{30}$$

On a fixed compact component subset the residual is $O_T(L^{-2})$, and (6) holds.

Proof. It remains to estimate the transition annuli. Put $n = \ell_i(N_p)$ and $\delta = \chi(\ell_i(K_i) - n)$. The curvature operator in logarithmic coordinates is a second-order expression with smooth coefficients on the compact matrix range. Compare its value at $n + \delta$ with its value at n, which is exactly zero. The difference is bounded by a constant times

$$|D^2\delta| + |Dn||D\delta| + |\delta|(|D^2n| + |Dn|^2) + |D\delta|^2. \tag{31}$$

Here $|Dn| \le C_T/(L\rho)$, $|D^2n| \le C_T/(L\rho^2)$, and each cutoff derivative of order k is $O(\rho^{-k})$. Applying (27) in (31) and integrating over area $O(\rho^2)$ gives

$$\int_{\text{transition}}|2i\Lambda F_{\widehat{H}_t}|\,dA_t \le C_T\left(\frac{\rho}{L} + \frac{(1+\log L)^2}{L^2}\right).$$

The logarithmic time derivative is bounded by C_T/L throughout the annulus, so its integral is at most $C_T\rho^2/L$. In particular it fits the same bound. Using the exactly flat profile in (31) is essential: a bound on the full interpolant Hessian would produce an order L^{-1} integral and would not suffice after time integration.

Sum these estimates over finitely many nodes and components, and use Lemmas 4.1 and 4.2. For $\rho = L^{-1}$ the exponentially small plumbing errors are absorbed in the right side of (29). Integration over the physical interval of length TL proves (30). A fixed compact component subset eventually lies entirely in a thick region. There the last assertion follows from (21) and Lemma 4.1. □

5. Tracking the actual metric flow

Proof of Theorem 1.1. The construction in Section 4 defines $\widehat{H}_t$ from the global finite trajectory. It gives uniform matrix bounds on each bounded slow-time interval and proves (6). Lemma 2.3 gives the actual solution for all physical times and preserves its initial bounds $cI \le H_t \le CI$. Thus actual and comparison metrics satisfy common two-sided bounds on $0 \le s \le TL$, independently of t.

Apply Lemma 2.2 to the two families and use Proposition 4.4. Uniformly for $0 \le s \le TL$,

$$\int_{X_t}\sigma(H_t(s), \widehat{H}_t(s))\,dA_t \le \eta_t + C_T\frac{(1+\log L)^2}{L}. \tag{32}$$

Since $d^2 \le \sigma$, this proves (7). The integration is on the entire smooth compact fibre; no boundary terms from truncated components have been omitted.

Fix $K_i \Subset U_i \Subset C_i \setminus \operatorname{Sing} X_0$. The transported base geometry is uniformly smooth and elliptic on U_i. For small t, the approximate residual there is $O_T(L^{-2})$. The nonnegative scalar function $u = \sigma(H_t, \widehat{H}_t)$ therefore satisfies

$$(\partial_s + \Delta_t)u \le C_T L^{-2}$$

on that region. Choose finitely many interior coordinate balls covering K_i, with doubled balls contained in U_i and radii at most $1/2$. All choices are independent of t.

For a backwards parabolic cylinder ending at $s_0 \geq 1$, the function $u + C_T L^{-2}(s_0 - s)$ is a nonnegative homogeneous subsolution. The local scalar parabolic mean-value inequality [SC92, Theorem 5.1], with exponent one, gives

$$\sup_{K_i(t)} u(s_0) \leq C_{K_i,T} \left(\int_{s_0-1}^{s_0} \int_{X_t} u \, dA_t \, ds + L^{-2} \right).$$

The ball radii and coefficient bounds determine the constant. Only fixed compact charts are used, so no global Sobolev constant of the degenerating fibre enters. The bound holds up to the upper time endpoint by continuity. Substituting (32) and then using (6) and the triangle inequality for d proves (8). □

Corollary 5.1. *Suppose $cI \leq H_t^{\mathrm{in}} \leq CI$ and, on every fixed compact component subset, the initial metric matrices converge in L^2 to H_i^0. Then the initial-error hypothesis of Theorem 1.1 holds. In particular initial derivatives may be unbounded as $t \to 0$.*

Proof. On a common compact range of positive matrices, σ is bounded by a constant times squared Euclidean matrix difference. On each fixed thick subset, $\widehat{H}_t(0)$ converges uniformly to H_i^0, so the integral of $\sigma(H_t^{\mathrm{in}}, \widehat{H}_t(0))$ there tends to zero. The trace function is uniformly bounded everywhere. The omitted node regions have area at most $C\rho_0^2$ for a fixed small outer cutoff ρ_0, by Lemma 3.1. First send $t \to 0$ for fixed ρ_0, then $\rho_0 \to 0$. □

The local estimate uses a fixed physical transient of length one; any fixed positive transient can be obtained by choosing smaller interior cylinders. On a fixed interval $\delta \leq \tau \leq T$ with $\delta > 0$, this transient is automatically passed for small t. The global estimate applies from the initial time. It allows bounded initial variation on the neck because its area tends to zero, but does not imply pointwise approximation of that variation at time zero.

The finite equation is the leading evolution on the specified time scale. The reconstructed metrics contain additional component and endpoint corrections needed to make their accumulated residual small. The argument does not identify the finite equation with an exact invariant-manifold restriction of HYM at fixed t, and does not replace finite T by an unbounded slow-time interval.

6. Small spectrum and rescaled linearized evolution

This section concerns the HYM linearization at the constant flat metric, and therefore the scalar Laplacian on matrix entries. It supplies an operator identification and uniform Hilbert-space estimates. It makes no assertion that a nonstationary metric has the same instantaneous spectrum.

Let $V = \mathbb{R}^N$ with inner product $\langle a, b \rangle_A = \sum_i A_i a_i b_i$. Define

$$q(a) = \sum_{p:\, i \leftrightarrow j} (a_i - a_j)^2, \qquad B = 2\pi A^{-1} Q,$$

where Q is the symmetric matrix of q. The component graph is connected, so its kernel consists of constant vectors. Write $0 = \mu_0 < \mu_1 \leq \cdots \leq \mu_{N-1}$ for the eigenvalues of $A^{-1}Q$. Let $0 = \lambda_0(t) < \lambda_1(t) \leq \cdots$ be the scalar fibre spectrum, counted with multiplicity, and let S_t be the span of the first N eigenfunctions.

Let $\Omega_i(r, t)$ denote the image under Lemma 3.1 of C_i with the discs of coordinate radius r about all its node punctures removed. Here r is fixed, positive, and smaller than the chosen

outer plumbing radius. In this section $o_r(1)$ denotes a quantity tending to zero as $t \to 0$ with r fixed.

We use the following scalar facts from [Wue26d]. They hold for the present induced Kähler metrics, and indeed under weaker hypotheses in that paper:

$$L\lambda_j(t) \longrightarrow 2\pi\mu_j \quad (0 \le j < N), \qquad \lambda_N(t) \ge \gamma > 0. \tag{33}$$

We also need the compactness and form estimates in its proof, rather than eigenvalues alone. For clarity we state precisely the consequences used here. On every fixed thick component subset, a unit vector in S_t differs in L^2 from its component average by $O_r(L^{-1/2})$. Its squared mass in fixed node neighbourhoods of radius r is at most $Cr^2 + o_r(1)$. These estimates hold uniformly for all combinations of the N eigenfunctions. Finally the capacitary lower bound gives

$$L\int_{X_t} |du|^2\, dA_t \ge 2\pi q(a(u)) - o_r(1)\|u\|^2 \qquad (u \in S_t), \tag{34}$$

where $a(u)$ consists of averages over $\Omega_i(r,t)$, using the same fixed coordinate radius as the capacitary cutoffs. In [Wue26d, Section 4] these statements are the thick estimates, the node-mass estimate, their extension to finite combinations, and the energy lower bound. The lower bound is obtained by pairing du with the differentials of logarithmic capacitary potentials. Thus it applies to an entire low subspace, not only to individual eigenfunctions.

Fix a sufficiently small outer radius r_0, and define

$$T_t : S_t \to V, \qquad (T_t u)_i = \frac{1}{\operatorname{area}\Omega_i(r_0,t)} \int_{\Omega_i(r_0,t)} u\, dA_t.$$

Adjoints in what follows use the actual fibre L^2 inner product and the weighted inner product on V.

Theorem 6.1. *For sufficiently small t, the averaging map T_t is invertible. The polar identification*

$$J_t = T_t^*(T_tT_t^*)^{-1/2} : V \longrightarrow S_t \tag{35}$$

is an isometry, and

$$\|J_t^* L\Delta_t J_t - B\|_{V\to V} \longrightarrow 0. \tag{36}$$

For every $T < \infty$,

$$\sup_{0\le\tau\le T} \|J_t^* e^{-L\tau\Delta_t} J_t - e^{-\tau B}\| \longrightarrow 0. \tag{37}$$

For $0 < \delta < T$, regarding J_t as an embedding into the full fibre L^2 space,

$$\sup_{\delta\le\tau\le T} \|e^{-L\tau\Delta_t} - J_t e^{-\tau B} J_t^*\|_{L^2\to L^2} \longrightarrow 0. \tag{38}$$

The shifted finite resolvents converge uniformly when their positive shift ranges over a compact interval. On $S_t^\perp$,

$$\|\Delta_t^{-1}\|_{L^2\to L^2} \le \gamma^{-1}, \qquad \|e^{-s\Delta_t}\|_{L^2\to L^2} \le e^{-\gamma s}. \tag{39}$$

Proof. First we prove that T_t is asymptotically isometric. Choose a fixed smaller radius $r < r_0$. The thick Poincaré estimate implies that the means at r and r_0 differ by $O_r(L^{-1/2})\|u\|$. On the thick pieces, the mass is the area times the squared mean, up to $o_r(1)\|u\|^2$. The

omitted mass is bounded by $Cr^2\|u\|^2+o_r(1)\|u\|^2$, and the omitted component area is $O(r^2)$. Uniformly on the unit sphere of S_t,

$$\left|\|T_tu\|_A^2-\|u\|^2\right|\le (Cr^2+o_r(1))\|u\|^2.$$

Let $t\to 0$ for fixed r and then $r\to 0$. This proves $T_t^*T_t=I+o(1)$. Both spaces have dimension N, so T_t is invertible for small t; (35) is an isometry and $T_tJ_t=(T_tT_t^*)^{1/2}\to I$.

Put $B_t=J_t^*L\Delta_tJ_t$. Use a fixed capacity radius compatible with r_0 in (34); changing between two fixed averaging radii only adds $O(L^{-1/2})\|u\|$ to the means. Substituting $u=J_ta$ gives, uniformly on V,

$$\langle B_ta,a\rangle_A\ge\langle Ba,a\rangle_A-\varepsilon_t\|a\|_A^2,\qquad \varepsilon_t\to 0.$$

The eigenvalues of B_t are $L\lambda_j(t)$ for $0\le j<N$. Equation (33) therefore gives $\operatorname{tr}(B_t-B)\to 0$. For the self-adjoint difference $D_t=B_t-B$,

$$\lambda_{\min}(D_t)\ge-\varepsilon_t,\qquad \lambda_{\max}(D_t)\le\operatorname{tr}D_t+(N-1)\varepsilon_t.$$

This proves (36), including when limiting eigenvalues are repeated. The form inequality is the information which makes this inference stronger than convergence of spectra.

Since S_t is a spectral subspace, $J_t^*e^{-L\tau\Delta_t}J_t=e^{-\tau B_t}$. Duhamel's formula for nonnegative self-adjoint matrices gives

$$\|e^{-\tau B_t}-e^{-\tau B}\|\le\tau\|B_t-B\|.$$

The resolvent identity similarly gives the bound $z^{-2}\|B_t-B\|$ for shifts $z>0$. The spectral gap in (33) proves (39). Decomposing the full fibre space into $S_t\oplus S_t^{\perp}$ bounds the difference in (38) by $T\|B_t-B\|+e^{-\gamma L\delta}$. □

The exclusion of $\tau=0$ in (38) is necessary: the identity on the orthogonal complement cannot disappear in operator norm at time zero. The polar identification need not carry the finite constant vector to an exactly constant function for each nonzero t; it does so asymptotically. No unshifted inverse on a space containing a kernel is asserted.

Let $\mathfrak{p}_r$ be the real space of trace-free Hermitian matrices, of dimension r^2-1. Tensoring Theorem 6.1 with its identity proves all corresponding HYM linearization statements. The finite equation linearized at $H_i=I$ is

$$A_i\partial_\tau u_i=2\pi\sum_{p:\,i\leftrightarrow j}(u_j-u_i),$$

which is exactly $-B\otimes I_{\mathfrak{p}_r}$. The estimates (39) are Hilbert-space inverse and heat bounds, not global unweighted higher-regularity estimates. The nonlinear proof instead uses Lemma 2.2 and the fixed-chart scalar mean-value inequality.

7. A noncommuting example and further questions

7.1. **The compact triangle.** Consider

$$\mathcal{X}=\{xyz-t(x^3+y^3+z^3)=0\}\subset\mathbb{P}^2\times\Delta. \tag{40}$$

Its central fibre is the reduced triangle of coordinate lines. At each vertex, the derivative of the defining equation with respect to t is nonzero; elsewhere on the central fibre one spatial derivative is nonzero. Hence the total space is smooth near the central fibre, and a smaller disc gives a smooth total space by properness.

A singular fibre at $t\neq 0$ would have $yz=3tx^2$, $xz=3ty^2$ and $xy=3tz^2$. No coordinate can vanish at such a projective point. Multiplication then gives $(3t)^3=1$. Thus the fibres with $0<|t|<1/3$ are smooth plane cubics and are connected curves of genus one. Restrict

the product Kähler form $\omega_{FS} + (i/2)dt \wedge d\bar{t}$, with projective lines of area one. Every central component has area one and every pair meets in one node. The graph matrix is

$$Q = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \qquad \operatorname{spec} Q = \{0, 3, 3\}. \tag{41}$$

Take the trivial rank-two bundle and initial component metrics H_1^0, H_2^0, H_3^0 from (9). Their determinants are one and they are positive. Direct multiplication gives

$$[H_2^0, H_3^0] = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} \neq 0. \tag{42}$$

If a single complex matrix g made all three congruences $g^* H_i^0 g$ diagonal, then $g^* g = D$ would be positive diagonal. Writing $g = UD^{1/2}$, with U unitary, would make both $U^* H_2^0 U$ and $U^* H_3^0 U$ diagonal. This contradicts (42).

The constructed initial metrics $\widehat{H}_t(0)$ are also not simultaneously diagonalizable by a fixed frame for small t. Choose one fixed point on each component and transport them to X_t. The three metric values tend to the displayed triple. For any positive triple (P, Q, R) that can be diagonalized by a common congruence, $P^{-1}Q$ and $P^{-1}R$ commute, since after the congruence they are simultaneously diagonal under similarity. Their commutator for the three transported values tends to (42), and so is nonzero for small t. Every holomorphic automorphism of the trivial bundle on a compact connected smooth fibre is a constant matrix, entrywise by the maximum principle. The obstruction therefore applies to fixed holomorphic frames on the fibre.

The finite triple remains noncommuting for a short positive slow-time interval by continuity. On any closed subinterval bounded away from zero, Theorem 1.1 transfers the same noncommutation test at the three points to the actual flow for small t. The example consequently contains noncommuting actual dynamics, not only a nonlinear notation for independent scalar equations. The trace-free Hermitian multiplicity is three. By (41), its six small positive linearized modes are each asymptotic to $6\pi/L$, and the remaining positive linearized spectrum has a uniform gap.

The matrices, their determinants and commutator, and the graph eigenvalues are checked by the accompanying exact symbolic computation. The displayed algebra supplies the proof; the computation is a reproducibility check.

7.2. **What the effective description retains.** The limiting small spectral subspace records componentwise parallel endomorphisms. Its mass is the sum of component areas times the matrix inner product. A neck measures the disagreement between its endpoint metrics, with coefficient fixed by its logarithmic conformal length. For finite amplitudes this disagreement is a Riemannian logarithm, rather than a difference of matrices in a chosen linear coordinate. The same geometric data therefore determine both the linearized operator and the nonlinear finite evolution.

There is a related finite-dimensional structure in the small two-form spectrum of nodal threefolds [Wue26c, Wue26e]. There the finite directions are cohomology classes lost under smoothing, the energy is built from exceptional-curve periods, and the mass is a harmonic norm. Those modes are ordinary two-forms. They are not automatically endomorphism modes of HYM metric flow. In particular a positive scalar gap can persist in those examples, so the trivial-bundle linearization need not inherit their small two-form eigenvalues.

The present theorem identifies a nonlinear evolution determined by the same geometric masses and capacities as the small linearized operator, in a setting where the componentwise parallel endomorphisms are explicit. It complements the fixed-background bundle-deformation results of [Wue26a, Wue26b], but does not make their regularity constants uniform on a degenerating base.

7.3. **Further directions.** The first extension is to nontrivial bundles and nonidentity gluing. Parallel endomorphisms on different normalization components must then be compared through the limiting connections and gluing maps. Their dimensions can differ, and the finite equation need not be the same unconstrained product flow. Verification of the actual bundle geometry and of all surviving modes is necessary.

A second question concerns simultaneous degeneration of the base and approach to a bundle stability wall. Neck capacities and small extension energies may become comparable. A common reduction would have to compute their coupling, rather than simply add independently derived graph operators. Higher-order node-smoothing rates also require separate geometric analysis: a local equation $uv = t^{\nu}$ with $\nu > 1$ does not have the smooth total space assumed here.

Finally, the finite-slow-time comparison does not determine the interchange of degeneration and infinite-time limits. Nor does it give a general HYM theorem on degenerating threefolds or reconstruct a smooth space from its eigenvalues. These questions require additional information about the bundle, local geometry and nonlinear estimates beyond the result proved here.

References

[Bha13] R. Bhatia, *The Riemannian mean of positive matrices*, in *Matrix Information Geometry*, F. Nielsen and R. Bhatia (eds.), Springer, 2013, pp. 35–51. doi:10.1007/978-3-642-30232-9_2.

[DW00] G. D. Daskalopoulos and R. A. Wentworth, *The Yang–Mills flow near the boundary of Teichmüller space*, Math. Ann. **318** (2000), no. 1, 1–42. doi:10.1007/s002080000098.

[Don85] S. K. Donaldson, *Anti self-dual Yang–Mills connections over complex algebraic surfaces and stable vector bundles*, Proc. London Math. Soc. (3) **50** (1985), no. 1, 1–26. doi:10.1112/plms/s3-50.1.1.

[HKKP18] F. Haiden, L. Katzarkov, M. Kontsevich and P. Pandit, *Iterated logarithms and gradient flows*, preprint, 2018. arXiv:1802.04123v1.

[Lam26] S. T. Lam, *Semistability and asymptotics of geometric flows*, preprint, 2026. arXiv:2606.09235v1.

[RTZ13] M. Rupflin, P. M. Topping and M. Zhu, *Asymptotics of the Teichmüller harmonic map flow*, Adv. Math. **244** (2013), 874–893. doi:10.1016/j.aim.2013.05.021.

[SC92] L. Saloff-Coste, *Uniformly elliptic operators on Riemannian manifolds*, J. Differential Geom. **36** (1992), no. 2, 417–450. doi:10.4310/jdg/1214448748.

[TAV13] R. Tron, B. Afsari and R. Vidal, *Riemannian consensus for manifolds with bounded curvature*, IEEE Trans. Automat. Control **58** (2013), no. 4, 921–934. doi:10.1109/TAC.2012.2225533.

[Wan08] Y. Wang, *Coupled vortex equations on complete Kähler manifolds*, Canad. Math. Bull. **51** (2008), no. 3, 467–480. doi:10.4153/CMB-2008-047-0.

[Wue26a] B. J. Wuebben, *Quantitative local models for Hermitian–Einstein metrics*, preprint, 7 November 2026. doi:10.5281/zenodo.24029090.

[Wue26b] B. J. Wuebben, *Uniform asymptotic phase for Hermitian–Yang–Mills flow near a polystable bundle*, preprint, 7 November 2026. doi:10.5281/zenodo.24038504.

[Wue26c] B. J. Wuebben, *Integral topology and small eigenvalues in Calabi–Yau conifold transitions*, preprint, 9 November 2026. doi:10.5281/zenodo.24096887.

[Wue26d] B. J. Wuebben, *Small eigenvalues on degenerating curves and the intersection form*, preprint, 10 November 2026. doi:10.5281/zenodo.24120959.

[Wue26e] B. J. Wuebben, *Small Hodge eigenvalues on nodal degenerations of threefolds*, preprint, 10 November 2026. doi:10.5281/zenodo.24120961.

[Zhu10] M. Zhu, *Harmonic maps from degenerating Riemann surfaces*, Math. Z. **264** (2010), no. 1, 63–85. doi:10.1007/s00209-008-0452-0.